

Lavanya Sai Bandla

Aspiring Software Engineer — Java Developer — AI & Cloud Enthusiast

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EDUCATION

Bachelor of Technology in Computer Science and Engineering
Velagapudi Ramakrishna Siddhartha Engineering College

2023 – 2027
CGPA: 8.99

TECHNICAL SKILLS

Programming Languages: Java, Python, C, JavaScript, SQL

Web Technologies: HTML, CSS, REST APIs

Databases: SQL

Tools: Git, GitHub, VS Code

Core Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Software Engineering

PROJECTS

Smart AI-IoT System for Real-Time Monitoring and Prediction of Psychological Disorders using Machine Learning

- Built an end-to-end IoT + AI system for real-time psychological stress detection using Arduino sensors (GSR, Piezo, PIR), Python, Flask AI API, Firebase cloud backend, and Next.js dashboard.
Used machine learning model- Random Forest and achieved 92.4% detection accuracy, real-time alerts (<250 ms latency), and secure Firebase-based authentication with live charts and automated buzzer control.
- **Technologies:** Arduino, Python, Flask, Firebase, Next.js, scikit-learn, IoT sensors

Gemini-Based YouTube Content Summarizer

- Developed an AI-powered YouTube Summarizer that automatically extracts, processes, and summarizes video transcripts using Google's Gemini 2.0 Flash model, achieving over 92% semantic accuracy.
- Implemented advanced NLP-based extractive and abstractive summarization techniques to convert long-form video content into concise summaries. Integrated YouTube Data API and built clean user interface using Streamlit.
- **Technologies:** Python, Gemini API, YouTube Data API, Streamlit, NLP libraries, Google Cloud

Java-Based Online Beauty Parlour Management System

- Developed a console-based Java application to manage beauty parlour operations including customer registration, service selection, appointment booking, billing, and feedback management.
- Designed modular components using Object-Oriented Programming principles including classes, inheritance, interfaces, encapsulation, and abstraction for better application structure.
- Implemented appointment scheduling with time-slot validation, payment processing, discount handling, and file-based data persistence for managing customer records.
- **Technologies:** Java, OOP, Collections, File Handling, Exception Handling

CERTIFICATIONS

- AWS Certified Cloud Practitioner - (Beginner)
- AWS Academy – Cloud Foundations, Machine Learning Foundations, Data Engineering
- Cisco – Programming Essentials in Java, Python, C, Cybersecurity
- NPTEL – Statistical Computing, Privacy and Security in Online Social Media
- edX – Python for Data Science

ACHIEVEMENTS

Amazon FFE-AFE Scholar

2023 – 2027

- Awarded for academic excellence and technical merit.